KTH Computer Science and Communication

Exam in DD2421 Machine Learning
2025-10-16, kl 14.00 – 18.00

Aids allowed: *calculator*, *language dictionary*.

To take this exam you must be registered to this specific exam as well as to the course.

In order to pass this exam, your score x first needs to be 20 or more (out of 42, full point). In addition, given your points y from the Programming Challenge (out of 18, full point), the requirements on the total points, $p = x + y$, are preliminarily set for different grades as:

$$
\begin{aligned}
54 < p \leq 60 &\rightarrow A \\
48 < p \leq 54 &\rightarrow B \\
42 < p \leq 48 &\rightarrow C \\
36 < p \leq 42 &\rightarrow D \\
29 < p \leq 36 &\rightarrow E \quad \text{(A pass is guaranteed with the required points for 'E'.)} \\
0 \leq p \leq 29 &\rightarrow F
\end{aligned}
$$

This exam consists of sections **A**, **B**, and **C**.
NB: use different papers (answer sheets) for different sections A, B, and C.
This means you shall start a new page after section A, and again after section B.

On the KTH exam submission paper (NOT this page), count each of our section-questions as one question for submission. So cross a box for A1, A2, A3, ..., B1, B2, B3, ..., C1, C2, C3, ...

A Graded problems

Potential inquiries to be addressed to Atsuto Maki.

A-1 Terminology (4p)

For each term (a–d) in the left list, find the explanation from the right list which *best* describes how the term is used in machine learning.

a) Curse of dimensionality

b) Bagging

c) RANSAC

d) Occams's razor

1) Problems in high computational cost

2) An example of ensemble learning

3) A concept of accepting high model complexity

4) Robust method to fit a model to data with outliers

5) A principle to choose the simplest explanation

6) Issues in data sparsity in space

7) An example of unsupervised learning

8) Random strategy for amplitude compensation

A-2 Nearest Neighbor, Classification (4p)

Suppose that we take a data set, divide it into training and test sets, and then try out two different classification procedures. We use *three-quarters* of the data for training, and the remaining *one-quarter* for testing. First we use *Logistic Regression* and get an error rate of 10% on the training data. We also get the average error rate (weighted average over both test and training data sets) of 12%. Next we use 1-nearest neighbor and get an average error rate (weighted average over both test and training data sets) of 6%.

a) What was the error rate with 1-nearest neighbor on the training set? (1p)

b) What was the error rate with 1-nearest neighbor on the test set? (1p)

c) What was the error rate with Logistic Regression on the test set? (1p)

d) Based on these results, indicate the method which we should prefer to use for classification of new observations, with a simple reasoning. (1p)

Briefly motivate your answer to each question.

A-3 Decision Trees/Forests and Entropy (6p)

a) Consider decision trees, and indicate a correct one as the basic strategy for selecting a question (attribute) at each node.

i. To minimize the expected reduction of the entropy.
ii. To maximize the expected reduction of the entropy.
iii. To minimize the expected reduction of gini impurity.

Simply indicate your choice. (1p)

b) Suppose we have trained a Decision Forest using five bootstrapped samples from a data set containing three classes, {Red, Yellow, Green}. We then applied the model to a specific test input, $\mathbf{x}$, and observed five estimates of class distributions from the five trees, respectively:

$\{P_1(\text{Red}|x), P_1(\text{Yellow}|x), P_1(\text{Green}|\mathbf{x})\}$ = {0.1, 0.5, 0.4}
$\{P_2(\text{Red}|x), P_2(\text{Yellow}|x), P_2(\text{Green}|\mathbf{x})\}$ = {0.2, 0.3, 0.5}
$\{P_3(\text{Red}|x), P_3(\text{Yellow}|x), P_3(\text{Green}|\mathbf{x})\}$ = {0.3, 0.6, 0.1}
$\{P_4(\text{Red}|x), P_4(\text{Yellow}|x), P_4(\text{Green}|\mathbf{x})\}$ = {0.2, 0.4, 0.4}
$\{P_5(\text{Red}|x), P_5(\text{Yellow}|x), P_5(\text{Green}|\mathbf{x})\}$ = {0.2, 0.3, 0.5}

Let us consider the average probability for combining these results into a single class prediction. What will the final class prediction be? *Motivate your answer by short phrases*. (2p)

c) Now imagine that you are playing with Cards and randomly sample *four cards* out of the pile of 52 cards (see Figure 1) *with replacement*, i.e. you sequentially draw a card but return it to the pile each time you have seen what it is.

At each instance of drawing a card, what is the Shannon information content of the outcome with respect to the suit, one of {*Spades, Hearts, Diamonds, and Clubs*}, measured in bits? Motivate your answer. (1p)

d) You play a game with a rule that you win if the suits of all the *four* cards are of *the same colour*, either *black or red*. Otherwise you lose. With respect to the outcome of this game, what is the expected information gain by drawing the first two card, i.e. by seeing (the suit colours of) the first card and the second card? Movivate your answer. (2p)

Note: if you do not have a calculator, answer with an expression but simplify it as much as possible.

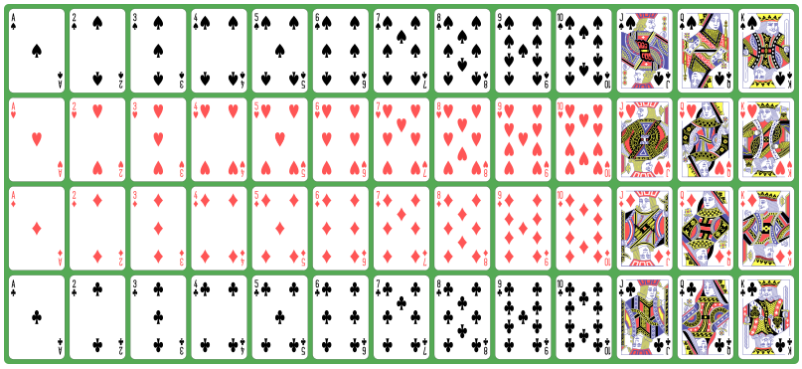


Figure 1. Playing cards consisting of 52 patterns.

A-4 Regression with regularization (4p)

For a set of N training samples $\{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\}$, each consisting of input vector $\mathbf{x}$ and output y, suppose we estimate the regression coefficients $\mathbf{w}^\top (\in \mathbf{R}^d) = \{w_1, \ldots, w_d\}$ in a linear regression model by minimizing

$$\sum_{n=1}^{N} (y_n - \mathbf{w}^\top \mathbf{x}_n)^2 + \lambda \sum_{i=1}^{d} |w_i|$$

with a value of λ for the regularization term.

a) What is this method called? (1p)

b) Now, let us consider a model with an extremely large λ and then models with gradually smaller values of λ (towards 0). The trainig error and test errors are then believed to change with respect to the bias-variance trade-off. Indicate which of {i,ii,iii,iv,v} is correct about the variance, and motivate your answer. The variance of the model will:
i. Remain constant.
ii. Steadily decrease.
iii. Steadily increase.
iv. Decrease initially, and then eventually start increasing in a U shape.
v. Increase initially, and then eventually start decreasing in an inverted U shape. (2p)

c) The method described is known to yield *sparse models*. Briefly explain what property of it enables the sparcity, in a short sentence. (1p)

A-5 PCA, Subspace Methods (4p)

Given a set of feature vectors which all belong to a specific class C (i.e. with an identical class label), we performed PCA on them and generated an orthonormal basis $\{\mathbf{u}_1, ..., \mathbf{u}_p\}$ which spans a p-dimensional subsapce, $\mathcal{L}$, as the outcome. Provide an answer to each question below.

a) What criterion are considered as useful for generating a basis in PCA?
i. Minimum squared distance criterion.
ii. Minimum variance criterion.
iii. Maximum squared distance criterion.
iv. Maximum variance criterion.
Simply choose what is/are valid. (2p)

b) Briefly explain what information can be referred for choosing an effective dimensionality of $\mathcal{L}$. (1p)

c) Now, we consider to solve a K-class classification problem with the Subspace Method and assume that a subspace $\mathcal{L}^{(j)}$ $(j = 1, ..., K)$ has been computed with training data for each class, respectively. Given a new input vector $\mathbf{x}$ whose class is unknown, we computed its projectiton length on each subspace as $S^{(j)}$ $(j = 1, ..., K)$, respectively. For a few classes among those with labels $\{l,m,n\}$, we had the following observations:
$S^{(l)}$ was the maximum of all $S^{(j)}$'s,
$S^{(m)}$ was the minimum of all $S^{(j)}$'s, and
$S^{(n)}$ was the closest to the average of all $S^{(j)}$'s.

Which class should $\mathbf{x}$ belong to? Simply choose a class label. (1p)

B Graded problems

Potential inquiries to be addressed to André Pereira.

B-1 Probability Theory (4p)

a) Deriving Bayes' Theorem (2p)
Which option(s) correctly show(s) the derivation of Bayes' theorem for $P(A \mid B)$?

a) The derivation starts from the sum rule, $P(A) = P(A, B) + P(A, \neg B)$, and rearranges the terms to isolate the posterior.

b) The derivation defines the joint probability in two ways, $P(A, B) = P(A \mid B)P(B)$ and $P(A, B) = P(B \mid A)P(A)$. It then sets them equal, $P(A \mid B)P(B) = P(B \mid A)P(A)$, and solves for $P(A \mid B)$.

c) The derivation assumes independence, stating $P(A, B) = P(A)P(B)$, and then divides by $P(B)$ to arrive at the theorem.

d) The derivation defines $P(A, B) = P(A \mid B)P(B)$ and then integrates over all possible values of A to find the marginal probability $P(B)$.

b) Combining the Chain Rule and Conditional Independence (2p)
The *chain rule of probability* states that for any n random variables $X_1, X_2, \ldots, X_n$,

$$P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P\big(X_i \mid X_1, X_2, \ldots, X_{i-1}\big).$$

The property of *conditional independence* means that, for some variables A, B, and C,

$$P(A \mid B, C) = P(A \mid C).$$

Using these two formulas, **rewrite the joint distribution** $P(A, B, C)$ so that it no longer depends on $P(A \mid B, C)$. Show the algebraic substitution clearly.

B-2 Maximum likelihood estimation for Regression (4p)

You wish to model a labeled dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ with the following model:

$$y = w_0 + w_1\, x^2 + \epsilon \tag{1}$$

where $\epsilon \sim \mathcal{N}(0, \sigma^2)$, i.e.,

$$P(\epsilon) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\epsilon^2/2\sigma^2}. \tag{2}$$

a) *Derive* a formula for the maximum likelihood estimate of w_1 as a function of w_0. *Show the steps of your derivations.* (3p)

b) In one sentence, *explain* what w_1 represents in this model. (1p)

B-3 Naïve Bayes Classification (4p)

You are given a small dataset for classifying emails as Spam ($y = 1$) or Not Spam ($y = 0$). The vocabulary consists of only two words: "prize" (x_1) and "meeting" (x_2). Each document is represented by a binary feature vector $x = [x_1, x_2]$, where $x_d = 1$ if the word d is present and 0 otherwise.

Using the training data, the following Maximum Likelihood Estimates were obtained:

$$\begin{aligned} P(y = 1) &= 0.5, & P(y = 0) &= 0.5, \\ P(x_1 = 1 \mid y = 1) &= 0.9, & P(x_2 = 1 \mid y = 1) &= 0.2, \\ P(x_1 = 1 \mid y = 0) &= 0.1, & P(x_2 = 1 \mid y = 0) &= 0.8. \end{aligned}$$

a) State the "naïve" assumption that this classifier makes. (1p)

b) A new email arrives containing both words, $x = [1, 1]$. Compute the unnormalised posterior probabilities $P(y, x)$ for both classes and decide the predicted label. (2p)

c) Explain how this final classification corresponds to the *Maximum A Posteriori (MAP)* decision rule. (1p)

C Graded problems

Potential inquiries to be addressed to Jörg Conradt.

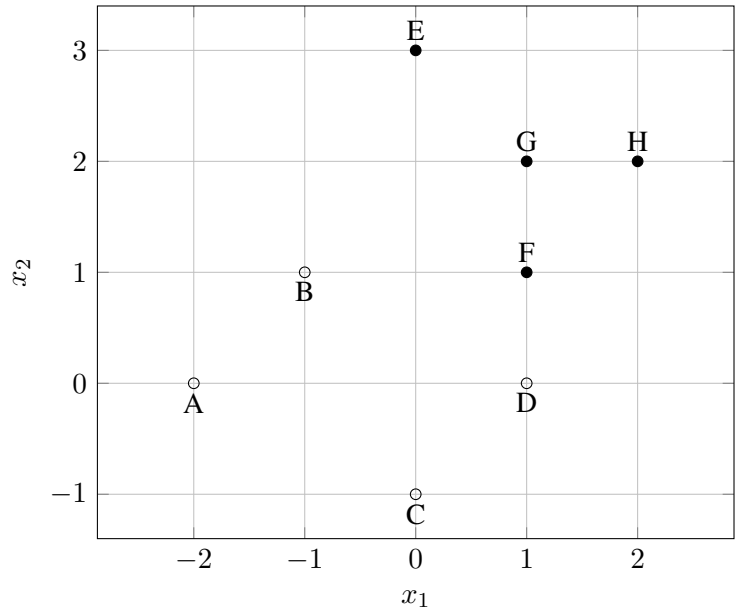
C-1 Warm-up: Support Vector Machine (1p)

Complete the following sentence: Out of all hyperplanes which solve a classification problem, the one with widest margin will probably ...

1. ... have the smallest number of parameters.
2. ... generalize best.
3. ... compute fastest.

Select one answer. Justify your answer with keywords.

C-2 Support Vector Machines (3p)



The eight datapoints in the figure are used to train a support vector machine. Filled circles are positive samples and unfilled circles are negative samples.

When a linear kernel ($\mathcal{K}(\vec{x}, \vec{y}) = (\vec{x} \cdot \vec{y} + 1)$) was used, and no slack was allowed ($C = \infty$), this resulted in three support vectors, with the corresponding α-values 0.5, 2, and 2.5.

To classify new points, the normal indicator function is used:

$$\text{ind}(\vec{x}) = \sum_k t_k \alpha_k \mathcal{K}(\vec{x}, \vec{s}_k) - b$$

a) What will the three support vectors be? You do not need to calculate this mathematically; use you knowledge about support vectors to deduce which three points must be support vectors. Justify **IN KEYWORDS or SHORT SENTENCES** (1p)

b) Which α-value must be associated with each of the three support vectors? **Give short motivations.** (2p)

C-3 Warm-up: Artificial Neural Networks (1p)

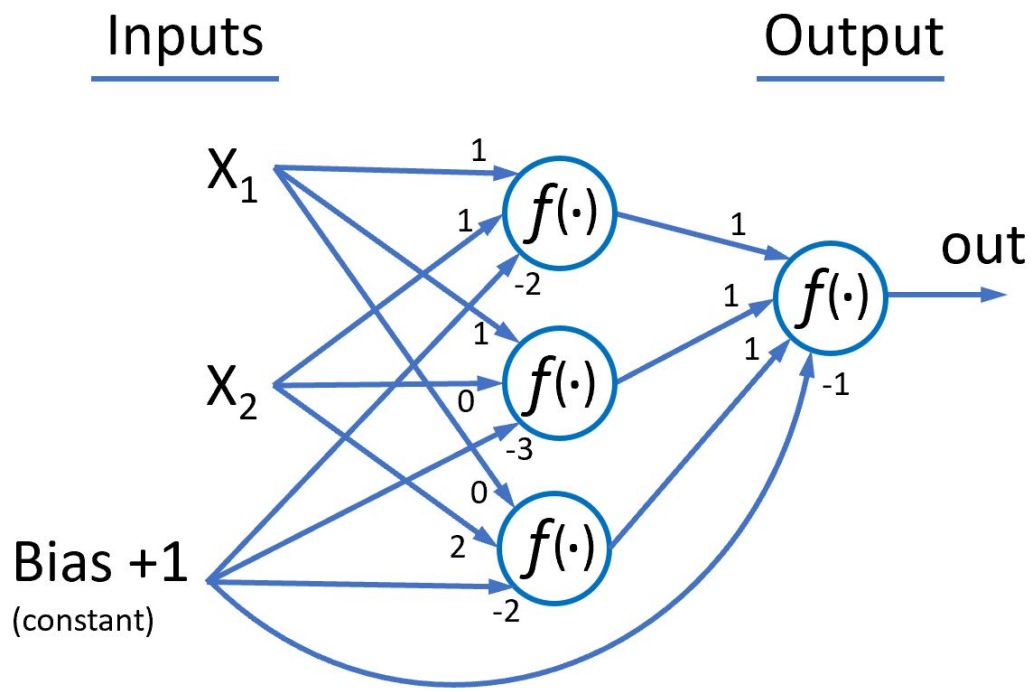
Select exactly **one** option of (1), (2), or (3), justify your answer **with keywords!**

Error-Backpropagation-Training in artificial neural networks mainly performs the following activity:

1. modifies all neurons' input weights and biases.
2. changes the connectivity between layers in the network.
3. identifies the best neuron activation function f(x).

C-4 Neuronal Networks (3p)

The following diagram shows a simple neuronal network with step activation functions in all neurons. The weight for incoming signals are shown directly at the neurons.



$f(\cdot)$ = step neuronal activation function

$$\text{step}\left(\sum_i (inp_i * wi)\right) = \begin{cases} 1 \text{ if } (\cdot) \geq 0 \\ 0 \text{ otherwise} \end{cases}$$

a) Draw a 2D diagram of the input space (two dimensional plot of X_1 vs. X_2) and show for which area of the input space the network produces a positive output. (2p)

b) Can this network be implemented in a single neuron with linear activation function (yes/no)? Explain **in KEYWORDS**. (1p)